

Course: ITAI 1371 – Intro to Machine Learning

Assignment: Module 02 Reflective Lab Journal

Group 1

Module 02 Reflective Journal: ML Development Tools

1. Core Technical Insights and Conceptual Breakthroughs

This reflective journal follows my experience completing the Module 02 Lab 02 for ITAI 1371, which focused on the essential development tools used in machine learning. While working through this lab, it showed how data is prepped and handled before it even runs through a machine learning model. Running the 16 cells showed me how beneficial tools like Pandas and NumPy are. On the surface, a Pandas data looks like a regular spreadsheet, but using a single line of code to clean and filter through data changes how you can handle a large dataset. This lab proved that data cleaning isn't just an extra step but allows for more accurate and refined data. My biggest takeaway was using NumPy vectorization instead of writing standard Python loops for math on a list of numbers. Using vectorization changes how you can run calculations on multiple matrices altogether. It removes the use of standard, unorganized loops, which is great for performance when working with larger datasets.

2. Challenges and Problem-Solving Strategies

The code within the 16 cells ran effortlessly without any execution errors. However, I ran into a problem when it came to turning a “.ipynb” into a PDF. When I tried to download the notebook, I received an error stating that it was unable to download due to not having Pandoc installed on my workstation. Instead of going through the process of downloading and setting up Pandoc to

download the notebook, I found a seamless solution. I was able to export the document as an HTML file, which perfectly retained all my data. When I opened the file in Microsoft Edge, I used the print command and was able to save it as a PDF. It taught me that working through errors and roadblocks is a huge part of development work.

3. New Perspectives and Future Applications

After finishing the lab and examining the data, it was a small glimpse into the day of a machine learning engineer's work. Before this module, it was not clear what all the jobs consist of.

Before, I used all I knew was working on neural networks and frameworks like PyTorch or TensorFlow. This assignment showed me how a huge portion of the job is data manipulation, preprocessing, and the clean-up of data using Pandas, NumPy, and other tools like Matplotlib.

Even when having to find workarounds for an error code is just as important as the coding.

Working through this lab from the initial code cells to the final environment workaround shows me more solutions for the problems I may encounter in the future.

4. Conclusion

Completing this lab gave me a better picture of the foundation on which machine learning is built on. Before the module, the tools felt like more applications that were downloaded and never used again. Knowing how to manipulate data, troubleshoot environment errors, and interpret visualizations is not just a prerequisite to machine learning; they are a core part. Going forward, I will use these tools to become more comfortable, so I am prepared for our machine learning model.